

KostalPiko 4.2 Reader

- Webscraping from KostalPiko Inverter
- MQTT publish
- Web interface for all
 - Wifi,
 - MQTT,
 - OTA &
 - KostalPiko inverter settings
- CSV and JSON file download



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1.1 KostalPiko 4.2 Reader Highlights

Functions list:

- Created for and only tested on a KostaPiko 4.2 Inverter
- Create at first start an Access Point for Wifi connection
- Let user connect device to their own home network. With static or DHCP IP number
- Read time from NTP server.
- Holds webserver with webpage to control and view status of the KostalPiko Reader
- Connect to MQTT to publish values
- Access to KostalPiko Reader is protected by a username and password.
- Reads KostalPiko values from website of KostalPiko inverter
- Webpage with KostalPiko values (all readonly)
- Webpage for setting wifi credentials
- Webpage for KostalPiko device credentials
- Webpage for setting MQTT credentials
- Use LED in flashing mode for special functions.
- Storage of KostalPiko device values in flash memory
- Overview of what is in onboard memory
- Storage for 128 days
- Use pushbutton for full reset
- Read KostalPiko values via CSV or JSON file
- Status of KostalPiko
- Programming via RS232 of OTA (password protected)

2 Naming

In this document two device are mentioned:

1. KostalPiko device (inverter). The device that is connected to the solar panels and converts the power from the panels to the power lines in your home. Although it has a web interface, automatic downloading and viewing in for example Home Assistant is not possible or difficult. Only KostalPiko 4.2 is supported.



Figure 1 Kostal-Piko device

2. KostalPiko reader. The reader reads the values from the KostalPiko device webpage and makes them visible through MQTT and the internal webpages. This device holds a ESP8266 processor and is connected via Wifi to your home network. Settings for the network connections can be made through the webpages of the KostalPiko device. See chapter 8 for more info.

3 Important information

- No internal battery in the KostalPiko Reader
- All settings and data are stored in internal flash memory or ESP8266. No storage in the cloud.
- No connections to external web sources. All HTML, CSS and Javascript files are inside the KostalPiko Reader.
- No information is sent to the external world. Everything is kept locally.

4 Hardware functionality of the KostalPiko Reader

The most important parts for the hardware of the KostalPiko reader are:

- ESP8266 processor. Holds all the software to read values from the KostalPiko device and makes them visible to the outside world. All Wifi, MQTT and KostalPiko device credentials can be set via the internal webpages.
- A flash memory device for storage of values from the KostalPiko device.
- Power supply.
- A button and a LED.

4.1 Button

- Press of 5 seconds on the button will reset the KostalPiko Reader completely. All network settings and the flash memory will be erased. This takes a maximum of 30 seconds. Reconnection to the reader via an Access Point.
- Button pressed are handled by interrupt.
- The LED is also used when the pushbutton is pressed. Every second the button is pressed, the LED will light for a short period, which makes counting easier.

4.2 LED

The LED will be used in several ways. During start-up of the KostalPiko, the hardware is tested for correct functionality. The next table shows how the LED is used.

Occurrence	Time on (sec)	Time off (sec)	Number of flashes	Remark
Power on	0.5	0	1	
File system LittleFS not found	1	5	Forever	This is the file system that is used to store all kind of settings from the KostalPiko. Without connection the KostalPiko Reader will not work correctly.
Wifi not found (wifi type = 0)	2	4	Forever	Without Wifi, NTP can also not be started. It is essential for the KostalPiko.
NTP server not found	3	3	Forever	If the time server cannot be reached, no data can be stored in the Flash memory, because no time stamps are available in Flash.
MQTT not found	4	2	3	MQTT can be used to send values to the MQTT server. Because a lot of people will not have MQTT, this is not a required feature.
Flash memory not found	5	1	Forever	Flash memory is used to store electrical value. If there is no free room anymore, a warning will be given. KostalPiko will work normally but will not store electrical values unless memory is freed.
Flash memory full	4	2	5	

When the software loop starts	1	1	2	Signals that the setup is done and the loop starts.
Reading the webpage of the KostalPiko (every minute)	0.5	--	1	Every time the KostalPiko device webpage is read

During the software startup a lot of connections and hardware parts are started. If a part cannot be started or is not function well, this can be seen as an error, because it stops some functions from the KostalPiko Reader.

5 Software functionality of the KostalPiko Reader

Software for the KostalPiko Reader is written in C++ and is created by using the Arduino IDE environment. A number of external libraries is used and they are defined in chapter 11.

5.1 Internal File system (LittleFS)

File system is used to store:

- html files (17 files)
- style.css file. All html files use the same style.css file (1 file)
- Icons for webpage (2 file)
- Files for login credentials (2 files)
- Files for Wifi credentials (8 files)
- Files for MQTT credentials (4 files)
- Files for KostalPiko device credentials (3 files)
- File for Ota password (1 file)

5.2 Flash chip

A large Flash memory (128Mbit/16MByte) is used to store KostalPiko values in memory. At a later time, these values can be read as a CSV or JSON file. See chapter 6.

5.3 Internet functionality

5.3.1 Internet/Wifi

Internet connection is used for:

- Starts with an access point (AP)
 - User can create connection to own network with dynamic IP
 - User can create connection to own network with static IP
- NTP server, to be able to store electrical value with time stamp
- Publish mqtt messages
- Webpages for the KostalPiko
- Reading CSV and/or JSON files with KostalPiko device values
- Settings credentials

5.3.2 Access Point

When the KostalPiko reader is started for the first time it creates an Access Point. Use a mobile device (laptop, telephone, tablet) to make a connection to this access point. The network that you should see is something like kostalpiko-xxxx, where xxxx is a 4 digit number, randomly generated between 1000 and 9999.

5.3.3 Login

The KostalPiko web pages are protected by a user login with “Username” and “Password”. Default both values are empty. The user can change and is advised to change this “Username” and “Password” to prevent unauthorized access.

5.3.4 Internet clock (NTP server)

Date and time are read from the NTP server and are used for timestamping KostalPiko values stored in flash memory. Date and time are also shown on the top of every webpage. Time zone is Amsterdam and fixed.

5.3.5 Webpages

All webpages are stored in processors memory of the KostalPiko Reader. Also, a CSS file is stored there to give every webpage the same look. In the webpages are so called Placeholders (terms as %....%). These terms are replaced by numbers/text or dates by a piece of software called “processor”, before the webpage is shown in a browser. In this way webpages are a little bit dynamic. Every 1 minute the date and time and sometimes the content of a webpage is refreshed.

5.3.6 MQTT

Via MQTT electrical values will be published to the MQTT broker. The following values will be sent:

- Current Power (W)
- Total Energy (kWh)
- Daily Energy(kWh)
- String1_voltage
- String1_current
- String2_voltage
- String2_current
- String3_voltage
- String3_current
- L1_voltage
- L1_power
- L2_voltage
- L2_power
- L3_voltage
- L3_power
- Status for the KostalPiko device (“MPP” or “noMPP”). MPP means delivering energy to the net.

5.3.7 System Settings

- Network name of the device. **Does not work for Static IP**

5.4 Software setup() and loop()

Using the Arduino IDE, there is a strict division between the setup and loop part. Both parts are described below.

5.4.1 Setup()

The following actions are performed here:

- Set serial connection and speed for debugging (serial) port
- Check if the littlefs file system exists
- If littlefs exists, read files for wifi and mqtt communication.

- Read files for login credentials
- Start the wifi connection depending of the files found in littlefs
- Start the webserver
- Start the timeserver
- Start the connection with the flash chip and find some free area to store electrical values
- Depending on the wifi type and the values for MQTT, start the MQTT connection.

At the end of the setup() routine the following flags are available

- `wifi_type`. What type of wifi connection do we have.
 - 0 = no wifi connection
 - 1 = Station mode with static IP
 - 2 = Station mode with dynamic IP
 - 3 = Access Point mode
- `mqtt_ok`. False if no connection can be made to the MQTT server or if `wifi_type` = 0.
- `ntp_ok`. False if no connection can be made to the time server or if `wifi_type` = 0 or 3.
- `littlefs_ok`. False if no file system can be found or files cannot be read.
- `flash_ok`. If flash chip is found and RDID and REMS values are correct.

Rules:

- If `wifi_type` = 0, there will be no other actions. No webserver, led will flash.
- If `wifi_type` = 3, no mqtt, no ntp and no flash actions will take place. Webserver will work.
- If `wifi_type` = 1 or 2, further checks will be done.
- If `littlefs_ok` = false, there will be no other actions. Led will flash.
- If `mqtt_ok` = false, no data will be sent to the mqtt broker. If "true" values will be sent to the MQTT broker
- If `ntp_ok` = false, there will be no other actions. Led will flash.

In general: Messages will be shown on a webpage, otherwise the LED will blink.

5.4.2 Loop()

Main loop runs on the external clock (ntp), if external time is available. Otherwise, the internal clock (millis) is used.

Main loop runs every minute and start with:

1. Connection to Wifi, ntp and MQTT is checked and if not available an attempt to reconnect is done. (still to implement)
2. If MQTT is active, values are published, Values are written into Flash memory
3. Led is driven
4. If erase flag is active a datagroup is erased.
5. If restart flag is active the KostalPiko is restarted.

Request for webpages and updates of them is done asynchronously on the background.

Buttons and led are processed by interrupts

5.5 KostalPiko value correction

The Daily Energy value received from the KostalPiko device is an increasing value during the day. This value is set back to zero, not at the end of the day, but at about 5:00 PM. This means that the Daily Energy value from yesterday will appear in the graph of the Daily Energy of today, which give a completely wrong graph for this Daily Energy.

In the KostalPiko Reader a new Daily Energy (out_daily_energy) value will be calculated. The value sent to the Flash memory and to MQTT will be set to zero at 00:00, so just when the new day starts. This value will be kept at zero until the KostalPiko Device starts to generate energy to the net. Blue is the original graph of Daily Energy and yellow the corrected one. Which is send by MQTT and saved in flash memory.

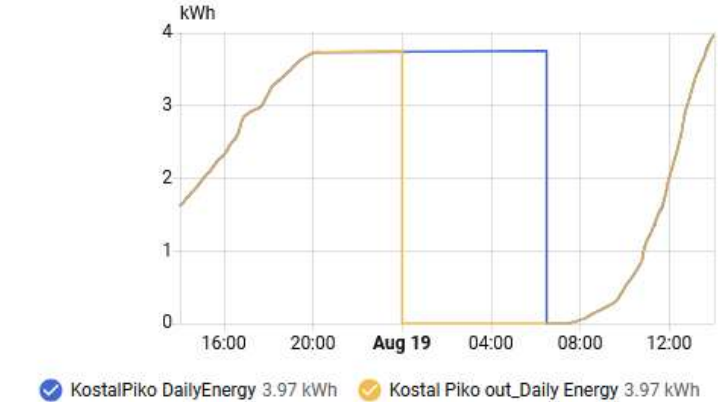


Figure 2. Modified Daily Energy

6 Usage of Flash memory

A flash memory is used (W25Q128 – 16 MByte) to store values from the KostalPiko Reader. If people do not have an MQTT server they can read the stored values at a later time, to process the data in their own way. Output formats will be JSON and CSV. No file system is used for the flash device.

Some numbers:

- A datagroup stores data of one day and has the size of 128kBytes.
- With a Flash size of 16Mbyte, data for 128 days can be stored.
- A dataset stores the data of one sample in the flash memory and the size of it is, 64 bytes, which give in one datagroup 2048 datasets. Storing data every minute in total one day has 1440 datasets.
- One dataset is used to store the year, month and date of the dataset itself.

6.1 User interaction with the stored data

On a webpage will be shown for which date, data is available in the flash memory. The user can select to download the data as CSV or JSON file. The user can also erase/delete the data. When the memory is full no more data will be stored in flash memory and new data will be lost. See Figure 6.

6.2 JSON file specification

Format of the example JSON file. Text must be between " " (not ") but numbers does not need them. Characters must be in [UTF-8](#) coding. Data is only available from the “on” time of the KostalPiko. Example file:

```
{
  "date": "2026-01-15",
  "format_version": "1.0",
  "samples": [
    {
      "time": "00:00",
      "flags": "0",
      "Current_AC_Power": "200",
      "Total_Energy": "12345",
      "Daily_Energy": "123.45",
      "Status": "MPP",
      "String1_voltage": "262",
      "String1_current": "0.64",
      "String2_voltage": "226",
      "String2_current": "0.65",
      "String3_voltage": "0",
      "String3_current": "0.0",
      "L1_voltage": "222",
      "L1_power": "154",
      "L2_voltage": "219",
      "L2_power": "0",
      "L3_voltage": "219",
      "L3_power": "129"
    },
    {
      "time": "00:00",
      "flags": "0",
```

```

    "Current_AC_Power":    "200",
    "Total_Energy":       "12345",
    "Daily_Energy":       "123.45"
    "Status":             "MPP",
    "String1_voltage":    "262",
    "String1_current":    "0.64",
    "String2_voltage":    "226",
    "String2_current":    "0.65",
    "String3_voltage":    "0",
    "String3_current":    "0.0",
    "L1_voltage":         "222",
    "L1_power":           "154",
    "L2_voltage":         "219",
    "L2_power":           "0",
    "L3_voltage":         "219",
    "L3_power":           "129"
  }
},
  "total_samples": 1440
}

```

The file downloaded will have the name <wifi network name>_<date>.json

Example: KostalPiko-7571_2026-01-16.json

6.3 CSV file specification

The file contains the following columns:

Time, Current_AC_Power(W), Total_Energy(kWh), Daily_Energy(kWh), Status(text), String1_voltage(V), String1_current(A), String2_voltage(V), String2_current(A), String3_voltage(V), String3_current(A), L1_voltage(V), L1_power(W), L2_voltage(V), L2_power(W), L3_voltage(V), L3_power(W), status

- Time is given as HH:MM.
- Flags is a number between 0 and 65535.

Example of the first part of a CSV file:

```

Time, Current_AC_Power(W), Total_Energy(kWh), Daily_Energy(kWh), String1_voltage(V),
String1_current(A), String2_voltage(V), String2_current(A), String3_voltage(V),
String3_current(A), L1_voltage(V), L1_power(W), L2_voltage(V), L2_power(W), L3_voltage(V),
L3_power(W), Status(text),
07:13, 17,60189, 0.00, 238, 0.08,215, 0.07,0, 0.00, 220,0,220,0,220,17,1
07:14, 20,60189, 0.00, 241, 0.09,214, 0.08,0, 0.00, 220,0,220,0,220,20,1
07:15, 20,60189, 0.00, 236, 0.09,215, 0.07,0, 0.00, 220,0,219,0,220,20,1
07:16, 19,60189, 0.00, 240, 0.09,206, 0.08,0, 0.00, 219,0,218,0,224,19,1
07:17, 20,60189, 0.00, 236, 0.09,203, 0.08,0, 0.00, 219,0,220,0,220,20,1
07:18, 18,60189, 0.00, 234, 0.09,202, 0.08,0, 0.00, 219,0,220,0,220,19,1

```

The file downloaded will have the name <wifi network name>_<date>.csv.

Example: KostalPiko-7571_2026-01-16.csv

6.3.1 During power-on

When the KostalPiko is plugged-in, it must be checked if there is a data-group active (having the same date as the actual date). There must be a valid time from NTP first.

- Copy all headers from flash to local memory for faster access
- Active data-group with same date found -> Continue using this data-group and find first unused memory location
- No data-group with same date -> Check if there is a free data-group.

- Write header with flag, year, month and date in flash and in header-array
- If no free data-group found, set full flag. Data will not be written to Flash anymore.

6.3.2 Actions during day change

Before electrical values are stored in flash, the date is checked. If the date of the data-group is not equal with the current date a new data-group must be started.

- A search starts for an unused data-group.
- If a free data-group is found, the header is written and values can be stored.
- If no free data-group is found, flash memory is full and no values will be written.
- Every 1 minute is will be checked if there is a free data-group.

6.3.3 Remove data of one day

If data from one day needs to removed, then 2 areas of 64Kbyte needs to be erased. Erasing one block of 64kbyte takes between 150 and 2 seconds. Erasing 2 of these areas will take 0.3 to 4 seconds. First the last 64kbyte block of a data-group is erased. Then the block before that. The last block is the block that holds the flag that indicates if this data-group is used or not.

7 Hardware

The block diagram of the design I used is shown below. If you have another way of supplying power to the design you can skip the AC_DC converter and maybe the AMS1117 voltage converter.

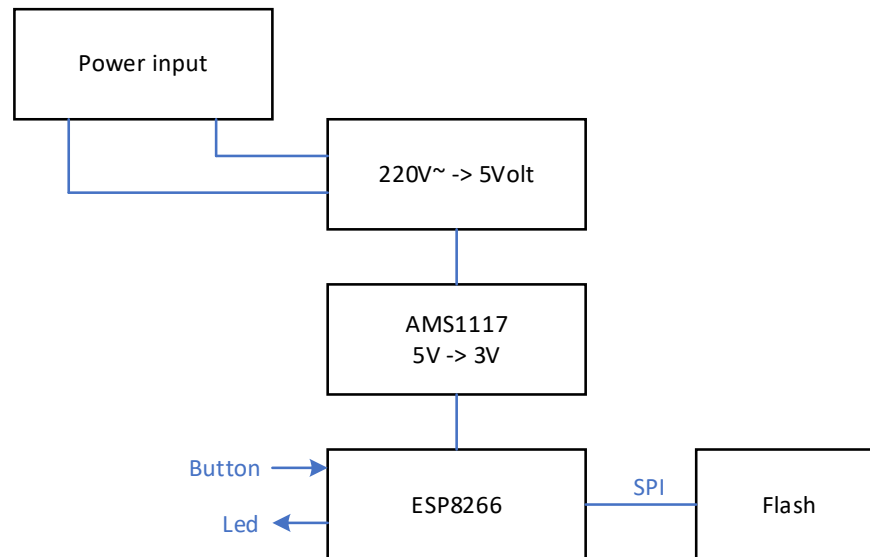


Figure 3. Block diagram of the electronics for the KostalPiko

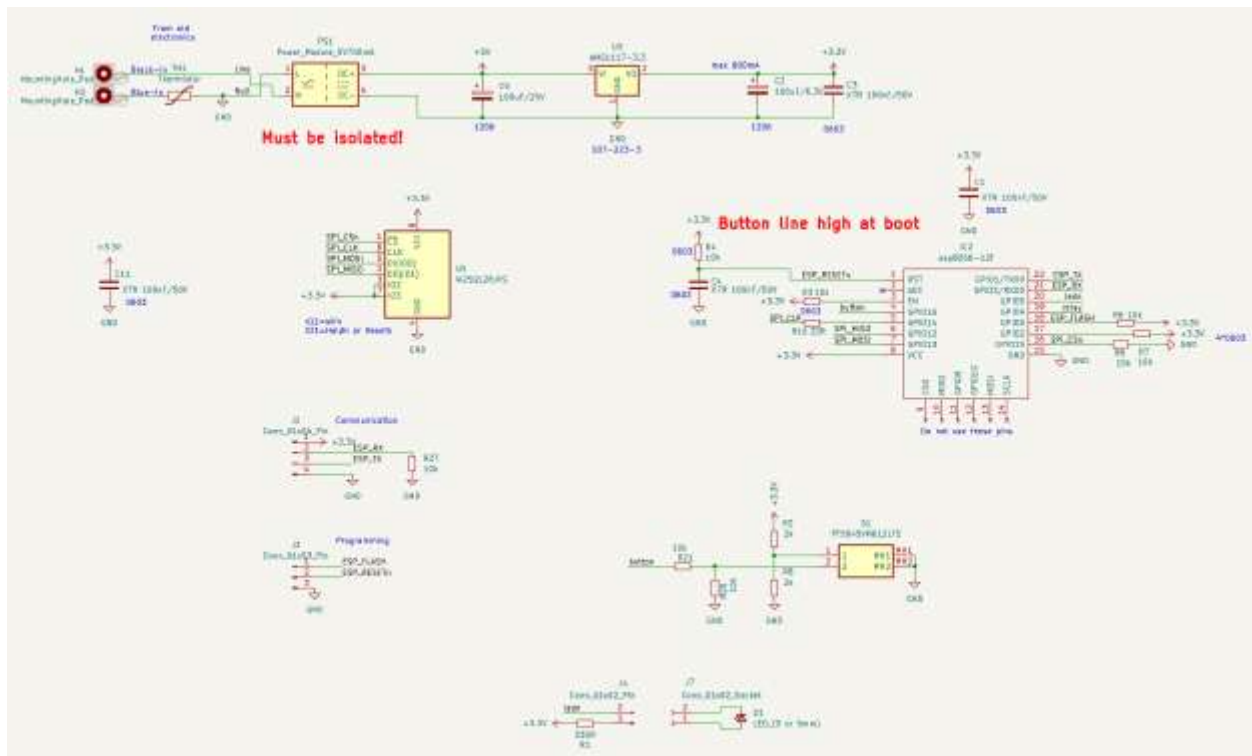


Figure 4. Schematic of the electronics for the KostalPiko

The schematic shows the connections to the ESP8266, Flash device, button and LED.

8 Webpages

Most of the settings for the KostalPiko Reader can be done by using the webpages. Also, information about already made settings is shown by the webpages.

Every web page has:

- Title specific for the page
- Green buttons that can be used to make a setting active or open another webpage
- Date and time are shown.
- Button to go back to the main page (except main page)
- Button to logout

The following webpages are created:

8.1 Main webpage (index.html)

This page starts when the IP number of the device is entered in the browser. First a login is needed to prevent unauthorized access of the KostalPiko Reader.

KostalPiko main page

Date: 2026/08/19 Time: 13:54:03

Selection	Meaning	Press the button
Kostal Piko values	View the electrical values from the KostalPiko.	KostalPiko Values
Memory	Gives an overview of what is the memory. Download datasets via CSV or JSON file from the memory. Also resetting datasets from the memory.	Memory
System Settings	Make system settings like network credentials or reset here.	System Settings
Last Compile Date	Jul 31 2026 17:41:57.	

Logout

Figure 5. Main webpage of the KostalPiko Reader

On this webpage some important settings can be found like:

- Link to KostalPiko energy values
- Link to memory filling and download/remove of values
- Link to system settings
- Last compile Date & Time.
- Logout button

The following subpages are defined and reachable by buttons on the main page.

8.2 KostalPiko_values

On this webpage values are shown from the KostalPiko device. The following values are shown:

- Actual/Current Power (W)
- Total Energy (kWh)
- Daily Energy (kWh)
- String1 Voltage (V)
- String1 Current (A)
- String2 Voltage (V)
- String2 Current (A)
- String3 Voltage (V)
- String3 Current (A)
- L1_voltage (V)
- L1_power (W)
- L2_voltage (V)
- L2_power (W)
- L3_voltage (V)
- L3_power (W)
- Status of the KostalPiko device

Kostal-Piko values
Date: 2026/05/25 Time: 12:10:55

Unit	Value
Actual Power (W)	3329
Total Energy (kWh)	58435.00
Daily Energy (kWh)	14.35
String1_Voltage (V)	226
String1_Current (A)	8.19
String2_Voltage (V)	205
String2_Current (A)	8.16
String3_Voltage (V)	0
String3_Current (A)	0.00
L1_Voltage (V)	229
L1_Power (W)	1128
L2_Voltage (V)	224
L2_Power (W)	1079
L3_Voltage (V)	232
L3_Power (W)	1121
Status	MPPT

[Back to Main page](#)

Figure 6. Kostal-Piko values

8.3 Memory

Shows a list of all the dates, of which values are available.

Three types of buttons are shown for every item.

- Download CSV file
- Download JSON file
- Delete dataset

Kostal-Piko Memory usage
Date: 2026/06/10 Time: 16:42:31

Area	Date	Get CSV	Get JSON	Delete
Memory Area 0	2026/6/4	Download CSV	Download JSON	Delete dataset
Memory Area 1	2026/6/5	Download CSV	Download JSON	Delete dataset
Memory Area 2	2026/6/6	Download CSV	Download JSON	Delete dataset
Memory Area 3	2026/6/7	Download CSV	Download JSON	Delete dataset
Memory Area 4	2026/6/8	Download CSV	Download JSON	Delete dataset

Figure 7. Memory filling of the Kostal-Piko

8.4 KostalPiko Settings webpage

On this webpage settings can be made for:

- Network credentials for Sign-in, Wifi, MQTT and KostalPiko device
- Reset Device

Kostal-Piko Settings page

Date: 2026/05/25 Time: 12:14:55

Selection	Meaning	Press the button
Network Credentials	View/Modify the Login, Wifi, MQTT and Kostal-Piko Credentials.	Kostal-Piko Reader network Credentials
Output control	See who is controlling the output.	Output Control
Reset	Reset Wifi, MQTT, Limits, Standbykiller, TimerSwitch values.	Reset

Back to Main page

Figure 8. Webpage for system settings

8.5 KostalPiko Reader network credentials

Via this webpage “KostalPiko Reader network Credentials”, Settings can be viewed and/or changed. Default view is with passwords replaced by ****. Use the “Toggle ..” button to view the passwords.

If the user wants to change settings for the credentials, then there are 4 buttons:

- Modify Sign-in Credentials
- Modify Wifi Credentials
- Modify MQTT Credentials
- Modify KostalPiko credentials

Using these buttons will open other webpages where the credentials can be changed. After saving the credentials, the KostalPiko will restart.

KostalPiko Login & Network Credentials

Date: 2026/08/19 Time: 16:38:55

Item	Value	Press the button
KostalPiko Reader Credentials		Modify Sign-in credentials
Login username:	admin	
Login Password:	*****	Toggle password
Wifi Credentials		Modify Wifi credentials
Wifi SSID:	Somnet	
Wifi Password:	*****	Toggle password
Wifi IP Address:		
Wifi Gateway:		
Wifi Subnet:		
Wifi Primary DNS:		
Wifi Secondary DNS:		
Wifi Network Name:	KostalPiko	
Wifi MAC:	9C:C7:A6:CE:7C:E3	
Wifi connection type:	Station mode with dynamic IP	
MQTT Credentials	MQTT connected	Modify MQTT credentials
MQTT Host:	192.168.2.160	
MQTT Port:	1883	
MQTT User:	mosquitto	
MQTT Password:	*****	Toggle password
KostalPiko Credentials	KostalPiko connection OK	Modify KostalPiko credentials
KostalPiko IP Address:	192.168.2.245	
KostalPiko User:	SN-Login	
KostalPiko Password:	*****	Toggle password
OTA Credentials		Modify OTA credentials
OTA Password:	*****	Toggle password

[Back to Main page](#)

To access the KostalPiko Device credentials are available.

For Wifi connection Wifi Credentials can be set

If a user does not have a MQTT broker running, these credentials can be left untouched.

Select Modify KostalPiko.. to get access to the KostalPiko device

OTA credentials are for a save way to update the software in the KostalPiko reader.

Figure 9. Showing network credentials

8.6 Reset

Kostal-Piko Reset page

Date: 2026/05/25 Time: 12:22:55

Selection	Meaning	Press the button
Reset Wifi Credentials	Removes the Wifi credentials from the Kostal-Piko Reader. The reader will restart and an Wifi Access Point is created. The current wifi connection to the Kostal-Piko Reader will be lost. Reconnect it to your network!	Reset Wifi
Reset MQTT Credentials	Removes the MQTT credentials from the Kostal-Piko Reader. Kostal-Piko Reader will restart!	Reset MQTT
Reset Kostal-Piko Credentials	Removes the Kostal-Piko credentials from the Kostal-Piko Reader. Kostal-Piko Reader will restart!	Reset KostalPiko
Reset Memory	All saved data from the memory will be removed. Kostal-Piko Reader will restart!	Reset Memory
Reset All Settings	Removes all settings from the Kostal-Piko Reader. Including the login for the Kostal-Piko Reader. Kostal-Piko Reader will restart!	Reset: All

[Back to Main page](#)

Figure 10. Reset several parts of the KostalPiko

The following selections for reset are possible:

- Reset Wifi credentials.
- Reset MQTT credentials.
- Reset all credentials.
- Reset Memory.
- Reset All Settings.

8.7 Login manager

Access to the KostalPiko Reader is protected by a username and password. This prevents unauthorized access to the KostalPiko Reader. At the start or after a full reset both terms are set to "admin". The user should change these. Simply fill in a new Username and Password and press "Submit Changes". The KostalPiko will restart, close the browser and open it again. Now the user can login with the new credentials.

Powerswitch Login Manager

Date: 2026/02/20 Time: 14:12:06

Item	Value
Username:	admin1
Password:	admin2
Submit:	Submit changes

[Back to Main page](#)

Figure 11. Changing the login credentials

8.8 WifiManager

This webpage can be used to set the Wifi credentials. There is a choice if the KostalPiko needs a fixed IP number of a dynamic IP number assigned by the router. This according to the following rules:

- If an IP number, gateway, subnet and PrimaryDNS are given, a connection with static IP number will be created.
- If an IP number, gateway or subnet are missing, a connection with dynamic IP will be created.
- For any other condition an Access Point will be created.

Wifi network name is limited to 30 characters

After these credentials are filled-in, the KostalPiko will restart and tries to connect to the network given in the credentials.

The type of connection that is created will be shown in the “System Settings” page

Keep in mind that the “Network Name” is used as device name on the internet and for for CSV and JSON download names.

Powerswitch Wi-Fi Manager
Date: 2026/02/28 Time: 16:24:40

Item	Value
SSID:	[REDACTED]
Password:	[REDACTED]
IP Address:	Fixed IP address for the Powerswitch
Gateway Address:	Mostly the IP address of your router
Subnet:	Probably 255.255.255.0
Primary DNS:	Probably the IP address of your router
Secondary DNS:	In most cases the IP address of your router
Network Name:	asdfghjklmnopqrstuvwxyz
Submit:	Submit changes

[Back to Main page](#)

Figure 12. Settings for Wifi

8.9 MQTTManager

If the user has an MQTT server running, connection can be made to this MQTT server. Credentials like: IP and Port number and also user and password need to be filled in. After filling-in the credentials the KostalPiko will restart and tries to connect to the MQTT server with the given credentials.

The settings for the MQTT connection will be shown in the “System Settings”-“KostalPiko network Credentials” page.

If the user does not have a MQTT server these items can be left blank.

Powerswitch MQTT Manager
Date: 2026/02/28 Time: 16:25:25

Item	Value
MQTT Host:	
MQTT Port:	1883
MQTT User:	
MQTT password:	
Submit:	Submit changes

Figure 13. MQTT settings

8.10 Webpage hierarchy

Webpage	Sub page		
Index (main)	KostalPiko_values		
	memory		
	system settings		
		show_wifi_mqtt_credentials	
			login
			wifimanager
			toggle_wifi_password
			mqttmanager
			toggle_mqtt_password
		reset	
			Reset Wifi/MQTT
			Reset Memory
			Reset All

Webpages are in html.

A “style.css” file is used to hold information about styles of the html pages.

A file “favicon.ico” that holds a small picture (16 x 16 pixels) that will be shown in the browser address line.

8.11 OTA programming

OTA programming is possible for the KostalPiko Reader

- Uploading firmware is protected by a password. Arduino IDE asks for the password.
- Uploading new datafiles to LittleFS has a problem. Arduino IDE does not ask for a password. A workaround for this can be the removal of the OTA password. Don't forget to add a password again.

9 ESP8266-12 (ESP8266MOD) pinning

Module pin no	Signal name	Alternative name	
1	RST	Used for reset button	ESP_RESET
2	ADC	ADC input	Not used
3	EN	Pulled high	High
4	GPIO16	High on boot (as output?) Used as input	Button
5	GPIO14	SPI-SCLK	Flash clock
6	GPIO12	SPI-MISO	Data to flash
7	GPIO13	SPI-MOSI	Data from flash
8	VCC	POWER	3.3Volt
9	CS0	GPIO11	Do not use them, they are used for the internal flash chip of the ESP8266 module.
10	MISO	GPIO7	
11	GPIO9	High on boot	
12	GPIO10	High on boot	
13	MOSI	GPIO8	
14	SCLK	GPIO6	
15	GND	GROUND	Ground
16	GPIO15	SPI-CS	CSn to flash
17	GPIO2	High on boot	
18	GPIO0	FLASH button	ESP_FLASH
19	GPIO4	n.c.	
20	GPIO5	Low will switch the LED on	Ledn
21	RXD0	GPIO3/high on boot	Debug-TX
22	TXD0	GPIO1/high on boot	Debug-RX

10 First Start of the KostalPiko Reader

When you first start with the KostalPiko Reader, the reader does not have any knowledge of your home network, KostalPiko Device and mqtt credentials.

10.1 Fully erase the KostalPiko Reader

To be sure that you start with a clean KostalPiko Reader:

- Plugin the device, wait until you see two flashes of the led. Now the Reader startup is completed.
- Press the button for 5 seconds. The LED will flash every second. Release the button after the fifth flash. The Reader will erase all settings and clears the flash memory.
- Wait for the restart of the Reader and the two LED flashes.

10.2 Connecting to your home network

- If the Reader is not connected to a network (does not have any credentials), it creates its own network.
- Use a mobile device (telephone, tablet or laptop) and look into the wifi networks. There must be a network called "KostalPiko_xxxx" (xxxx = a random number). Connect to this network.
- Now go with your webbrowser to <http://192.168.4.1/wifimanager.html>
- A login username and password are asked. They are not set, so click "Signin".
- You will get a webpage as shown on the right side.
- Fill-in at least your SSID, Password and Network name if you want to give the Reader a dynamic IP address.
- If you want to give the Reader a static IP address, you also have to fill-in the other items.
- Press "Submit changes" when you are done and want to use the given values.
- Press "Back to Main page" if you don't want to make changes to the Wifi settings.
- When you submit the changes, the values will be saved and the Reader will restart and will connect to your home network.
- Find your Reader now in your own network, by entering the IP address of your Reader in your internet browser.
- From the main page go to page "System Settings" & "Kostal-Piko Reader network Credentials"
- You will see that there is no connection to the KostalPiko device (red text).

KostalPiko Wi-Fi Manager
Date: 2026/06/20 Time: 11:33:43

Item	Value
SSID:	██████
Password:	██████████
IP Address:	Fixed IP address for the Kostal_Piko Reader
Gateway Address:	Mostly the IP address of your router
Subnet:	Probably 255.255.255.0
Primary DNS:	Probably the IP address of your router
Secondary DNS:	In most cases the IP address of your router
Network Name:	██████
Submit:	Submit changes

Figure 14. Setting Wifi credentials

Kostal-Piko Credentials	KostalPiko connection NOK	Modify Kostal-Piko credentials
--------------------------------	----------------------------------	---

- Click on “Modify KostalPiko credentials”.
- Now you can enter the IP address of your KostalPiko device, a username and a password. See the next picture
- If OK click on “Submit changes”.
- Again, the Reader will restart will try to connect to the KostalPiko device
- Now go to the main page of the website of the Reader in select “System Settings” & “Kostal-Piko Reader network Credentials”.
- Check the KostalPiko credentials and you should see in green “KostalPiko connection OK”
- If you see text in red again, then the connection is not successful.

KostalPiko Access Manager
Date: 2026/06/20 Time: 11:47:47

Item	Value
Kostal-Piko IP-Address:	
Kostal-Piko User:	
Kostal-Piko password:	
Submit:	Submit changes

Figure 15. Setting KostalPiko credentials

Kostal-Piko Credentials	KostalPiko connection OK	Modify Kostal-Piko credentials
--------------------------------	---------------------------------	---

Figure 17. Connection to KostalPiko device

- If you have an MQTT server running, you can set the MQTT credentials in the same way.
 - If don't have an MQTT server running or don't know what MQTT is, leave these fields untouched
 - Picture on the right show what values are published
-
- To prevent unauthorized access to the KostalPiko Reader you can add a login username and password.
 - The KostalPiko Reader can be reprogrammed over the air (OTA). To prevent unauthorized reprogramming a password can be set via “OTA Password”.

MQTT Explorer

Application Edit View

MQTT Explorer Search...

▼ 192.168.2.160

- ▶ p1-meter (16 topics, 32 messages)
- ▶ homeassistant (14 topics, 14 messages)
- ▶ esphome (1 topic, 1 message)
- ▶ evcc (77 topics, 77 messages)
- ▼ KostalPiko
 - ▼ sensor
 - status = MPP
 - Current_Power = 3237
 - Total_Energy = 58935
 - Daily_Energy = 12.83
 - String1_Voltage = 231
 - String1_Current = 7.85
 - String2_Voltage = 205
 - String2_Current = 7.89
 - String3_Voltage = 0
 - String3_Current = 0.00
 - L1_output_voltage = 222
 - L1_output_power = 1057
 - L2_output_voltage = 224
 - L2_output_power = 1060
 - L3_output_voltage = 231
 - L3_output_power = 1119

Figure 16. Published MQTT values

11 YAML file settings for Home Assistant

Under

mqtt:

sensor:

place

```
#
# Kostal Piko Solar inverter
#
# Power & Energy
- name: "Current_Power"
  state_topic: "KostalPiko/sensor/Current_Power/state"
  unique_id: "KP_CurPow"
  unit_of_measurement: W
  device_class: "power"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: Total_Energy
  state_topic: "KostalPiko/sensor/Total_Energy/state"
  unique_id: "KP_TotEne"
  unit_of_measurement: kWh
  device_class: "energy"
  state_class: "total_increasing"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: "Daily Energy"
  state_topic: "KostalPiko/sensor/Daily_Energy/state"
  unique_id: "KP_DaiEne"
  unit_of_measurement: kWh
  device_class: "energy"
  state_class: "total"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
# string1
- name: "String1_Voltage"
  state_topic: "KostalPiko/sensor/String1_Voltage/state"
  unique_id: "KP_Str1_V"
  unit_of_measurement: V
  device_class: "voltage"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: "String1_Current"
  state_topic: "KostalPiko/sensor/String1_Current/state"
  unique_id: "KP_Str1_A"
  unit_of_measurement: A
```

```

device_class: "current"
state_class: "measurement"
"device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
}
#string2
- name: "String2_Voltage"
  state_topic: "KostalPiko/sensor/String2_Voltage/state"
  unique_id: "KP_Str2_V"
  unit_of_measurement: V
  device_class: "voltage"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: "String2_Current"
  state_topic: "KostalPiko/sensor/String2_Current/state"
  unique_id: "KP_Str2_A"
  unit_of_measurement: A
  device_class: "current"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
# L1
- name: "L1_output_Voltage"
  state_topic: "KostalPiko/sensor/L1_output_voltage/state"
  unique_id: "KP_L1_Volt"
  unit_of_measurement: V
  device_class: "voltage"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: "L1_output_Power"
  state_topic: "KostalPiko/sensor/L1_output_power/state"
  unique_id: "KP_L1_Pow"
  unit_of_measurement: W
  device_class: "power"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
# L2
- name: "L2_output_Voltage"
  state_topic: "KostalPiko/sensor/L2_output_voltage/state"
  unique_id: "KP_L2_Volt"
  unit_of_measurement: V
  device_class: "voltage"
  state_class: "measurement"
  "device": {

```

```

        "identifiers": ["kostalpiko"],
        "name": "Kostal Piko",
        "model": "Piko",
        "manufacturer": "Kostal"
    }
- name: "L2_output_Power"
  state_topic: "KostalPiko/sensor/L2_output_power/state"
  unique_id: "KP_L2_Pow"
  unit_of_measurement: W
  device_class: "power"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}
#L3
- name: "L3_output_Voltage"
  state_topic: "KostalPiko/sensor/L3_output_voltage/state"
  unique_id: "KP_L3_Volt"
  unit_of_measurement: V
  device_class: "voltage"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}
- name: "L3_output_Power"
  state_topic: "KostalPiko/sensor/L3_output_power/state"
  unique_id: "KP_L3_Pow"
  unit_of_measurement: W
  device_class: "power"
  state_class: "measurement"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}
# status
- name: "Kostal-Piko_Status"
  state_topic: "KostalPiko/sensor/status/state"
  unique_id: "KP_status"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}
# http code
- name: "Kostal-Piko_code"
  state_topic: "KostalPiko/sensor/kostalpiko_code/state"
  unique_id: "KP_code"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}
- name: base_Total_Energy
  state_topic: "KostalPiko/sensor/base_Total_Energy/state"
  unique_id: "KP_Base_TotEne"
  unit_of_measurement: kWh

```

```

device_class: "energy"
state_class: "total_increasing"
"device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
}
- name: out_Total_Energy
  state_topic: "KostalPiko/sensor/out_Total_Energy/state"
  unique_id: "KP_out_TotEne"
  unit_of_measurement: kWh
  device_class: "energy"
  state_class: "total_increasing"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
- name: "out_Daily_Energy"
  state_topic: "KostalPiko/sensor/out_Daily_Energy/state"
  unique_id: "KP_out_DaiEne"
  unit_of_measurement: kWh
  device_class: "energy"
  state_class: "total"
  "device": {
    "identifiers": ["kostalpiko"],
    "name": "Kostal Piko",
    "model": "Piko",
    "manufacturer": "Kostal"
  }
}

```

12 Values in Home-assistant

I have created an item under Overview with all the KostalPiko values



Figure 18. KostalPiko values in Home Assistant

Also created a link for direct access to the KostalPiko Reader website

Home Assistant

Kostalpiko

Overview

KostalPiko

Map

Energy

Activity

History

HACS

Media

KostalPiko main page

Date: 2026/09/28 Time: 16:27:34

Selection	Meaning	Press the button
Kostal Piko values	View the electrical values from the KostalPiko.	KostalPiko Values
Memory	Gives an overview of what is the memory. Download datasets via CSV or JSON file from the memory. Also resetting datasets from the memory.	Memory
System Settings	Make system settings like network credentials or reset here.	System Settings
Last Compile Date	Aug 19 2026 16:35:50.	

[Logout](#)

Figure 19. Direct access to KostalPiko Reader website

13 Used libraries/plugin's in Arduino IDE

- AsyncMqttClient, Marvon Roger
<https://github.com/marvinroger/async-mqtt-client>
- Ethernet, Arduino
<https://www.arduino.cc/en/Reference/Ethernet>
- ESP_Async_Webserver, Mathieu Carbou
<https://github.com/mathieucarbou/ESPAsyncWebServer>
- ESP_AsyncTCP, me-no-dev
<https://github.com/me-no-dev/ESPAsyncTCP>
- FloattoAscii, Koen de Vleeshauwer
<https://github.com/koendv/FloatToAscii>
- NTPClient, Fabrice Weinberg
<https://github.com/arduino-libraries/NTPClient>
- Time, Paul Stoffregen (Arduino)
<https://playground.arduino.cc/Code/Time/>
- Timezone, Jack Christensen
<https://github.com/JChristensen/Timezone>
- EspSoftwareSerial, Dirk Kaar
<https://github.com/plerup/espsoftwareserial/>
- SPIMemory
<https://github.com/Marzogh/SPIMemory>

13.1 Littlefs upload, For uploading HTML, CSS, JS and text files

Download file: arduino-littlefs-upload-1.6.2.vsix

Place in: C:\Users\<user>\.arduinoIDE\plugins

In Arduino use CTRL-SHIFT-P and choose: "Upload Littlefs.."

Serial Monitor must be switched off

Use flash and reset button on the ESP module if needed.

13.2 ESP Exception decoder

For debugging:

- EspExceptionDecoder
<https://github.com/me-no-dev/EspExceptionDecoder>